Practical exercise 5

Linear regression.

1. Two scientists, A and B, are investigating the relationship between two variables, under the usual normal linear regression model, with $E[Y|x] = a + bx$ and error variance σ^2

   Scientist A designs her experiment by observing the response variable Y at the eight values of x given by $x = 1, 2, 3, 4, 5, 6, 7$, and 8. Her results are:

   ```
   x:  1    2    3    4    5    6     7    8
   y: 15.4 12.8 18.5 16.5 19.4 23.4 18.4 26.0
   ```

   Scientist B designs his experiment by observing Y at the eight values of x given by $x = 1, 1, 1, 1, 8, 8, 8$, and 8. His results are:

   ```
   x:  1    1    1    1    8    8     8    8
   y: 15.3 10.8 14.5 11.5 25.5 26.9 20.3 26.0
   ```

   (a) Enter the data. Suggested object names: `xa, ya, xb, yb`

   (b) Analyse the results of both scientists' experiments.
   Your analysis in each case should include a scatterplot, parameter estimates $\hat{a}, \hat{b}$ and $\widehat{\sigma^2}$, the coefficient of determination R^2, the standard errors of the parameter estimates and appropriate comments.

   (c) State, with reasons, which experiment you think is better as regards the estimation of the slope b, and comment briefly on how the designs of the experiments influence this.

   (d) Produce plots for each experiment and have the computer add the best straight line through the points.

   (e) Produce residual plots for each experiment and discuss the graphs.

2. Obtain and attach the data set `cats` by typing the following:

```
library(MASS)
data(cats)
attach(cats)
```

Have a look at the data by typing `cats` and also use the `?cats` command.

In what follows we will explore the dependence of Heart Weight (`Hwt`) on Body Weight (`Bwt`). Plot Heart Weight against Body Weight in the usual way.

Fit a linear regression model to these data and add the estimated straight line through the points using the `abline` command.

Now obtain a plot of the residuals against `Bwt`. Also obtain a plot of residuals against fitted values of `Hwt`. What do these plots show? (For a good fit, look for spread around the line $y = 0$ and no obvious trend.)

Finally for this example, try the commands:

```
result=lm(Hwt~Bwt)
par(mfrow=c(2,2))
plot(result, which=1:4)
```

Return the graphics window to normal using

```
par(mfrow=c(1,1))
```

So far, the analysis has not looked at all at the gender of the cat.

Type in the `R` code

```
boxplot(Hwt ~ Sex, xlab='gender', ylab='heart weight')
```

which clearly shows that heart weight depends on gender.

Similarly, the `R` code

```
plot(Hwt~Bwt, xlab="Body weight (kg)", ylab="Heart weight (g)", type="n")
points(Hwt[Sex=="M"]~Bwt[Sex=="M"], pch=1, col="blue")
points(Hwt[Sex=="F"]~Bwt[Sex=="F"], pch=3, col="red")
legend(x="topleft",pch=c(1,3),legend=c("Male","Female"),
       col=c("blue", "red"), cex=0.8)
```

produces a plot of heart weight against body weight for male cats (denoted by o) and female cats (denoted by +).

3. The `R` data frame `January` is available on the course web page on Canvas.

   The data are the maximum January temperatures (degrees Fahrenheit), over the period 1931–1960 for 40 selected cities in the US, together with the latitude, longitude, and altitude (feet) of each.

   We treat temperature as the response variable and are interested in modelling its dependence on the explanatory variables latitude and longitude (for this exercise we do not consider altitude). In doing so it is important to use our physical knowledge of how temperature might be expected to depend on these variables, as well as the data given.

   (a) Give plots of temperature against latitude and temperature against longitude. In each case, comment on any dependence observable in the scatter plot. Identify (by name) and discuss briefly any anomalous observations.

   Note: You can label points on a plot by the following method:

   `identify(temperature ~ latitude,labels=row.names(january))`

   Then, move your mouse over the plot until you reach the point you wish to identify. Click the (left) mouse button and the name will appear. Repeat as desired. To get out of this mode you need to press the RIGHT mouse button and select `stop`.

   Construct also a plot of longitude against latitude and comment.

   (b) Use least squares regression to fit a linear relationship

   `temperature` $= a + b \times$ `latitude` $+ \epsilon$

   Give the estimated values of the coefficients a and b , and display a plot of temperature against latitude showing the fitted line. Construct and give also a plot of the residuals from the fitted model against latitude. Discuss the success of the fitted model.

   (c) Discuss briefly whether it might it be a good idea to exclude from further analysis the observation corresponding to Juneau, Alaska (observation 3).

   (d) Refit the linear relationship omitting from the analysis the observation corresponding to Juneau, Alaska. Again give the (new) estimated values of the coefficients a and b, and again display a further plot of temperature against latitude showing the refitted line.

   `regress2=lm(temperature[-3] ~ latitude[-3])`, etc.

   Construct a plot of the residuals from the refitted model against latitude. By comparing these plots with those obtained for the earlier analysis, comment on the improvement, or otherwise, in the fitted relationship.

   (e) Now give a plot of the residuals from the (new) fit of the model against the second explanatory variable longitude. Comment on the observed pattern and on the likely dependence on longitude of the original response variable `temperature`.

   (f) Would you expect either of the fitted models to be valid outside the continental USA? Explain briefly.